progress that has been made over the last two decades, we still have far to go before we can claim to have achieved full mastery.

And so string theory should be viewed as a work in progress whose partial completion has already revealed astonishing insights into the nature of space, time, and matter. The harmonious union of general relativity and quantum mechanics is a major success. Furthermore, unlike any previous theory, string theory has the capacity to answer primordial questions having to do with nature's most fundamental constituents and forces. Of equal importance, although somewhat harder to convey, is the remarkable elegance of both the answers and the framework for answers that string theory proposes. For instance, in string theory many aspects of nature that might appear to be arbitrary technical details—such as the number of distinct fundamental particle ingredients and their respective properties—are found to arise from essential and tangible aspects of the geometry of the universe. If string theory is right, the microscopic fabric of our universe is a richly intertwined multidimensional labyrinth within which the strings of the universe endlessly twist and vibrate, rhythmically beating out the laws of the cosmos. Far from being accidental details, the properties of nature's basic building blocks are deeply entwined with the fabric of space and time.

In the final analysis, though, nothing is a substitute for definitive, testable predictions that can determine whether string theory has truly lifted the veil of mystery hiding the deepest truths of our universe. It may be some time before our level of comprehension has reached sufficient depth to achieve this aim, although, as we will discuss in Chapter 9, experimental tests could provide strong circumstantial support for string theory within the next ten years or so. Moreover, in Chapter 13 we will see that string theory has recently solved a central puzzle concerning black holes, associated with the so-called Bekenstein-Hawking entropy, that has stubbornly resisted resolution by more conventional means for more than twenty-five years. This success has convinced many that string theory is in the process of giving us our deepest understanding of how the universe works.

Edward Witten, one of the pioneers and leading experts in string theory, summarizes the situation by saying that "string theory is a part of twenty-first-century physics that fell by chance into the twentieth